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September 12, 2026 · 2 Day Project

Featured Project: Secure Cloud Video Access Control (AWS Proof of Concept)

A fully serverless, identity-aware video delivery system built end-to-end on AWS, designed and implemented independently to evaluate a secure alternative to third-party private video hosting for internal/restricted content.

What I Built

- **Private storage:** Video hosted in a locked-down Amazon S3 bucket with all public access blocked.
- **Secure delivery:** Amazon CloudFront serves content via Origin Access Control. S3 is unreachable by any other path.
- **Cryptographic access gate:** CloudFront requires a signed cookie, issued only after identity verification.
- **Identity & authorization:** Amazon Cognito manages login and group membership (approved vs. not approved).
- **Custom authorization logic:** AWS Lambda checks group membership and signs time-limited access on the fly.
- **One-click experience:** API Gateway + a lightweight login flow. The user clicks one link, logs in, and is automatically granted or denied.

Outcome & Business Value

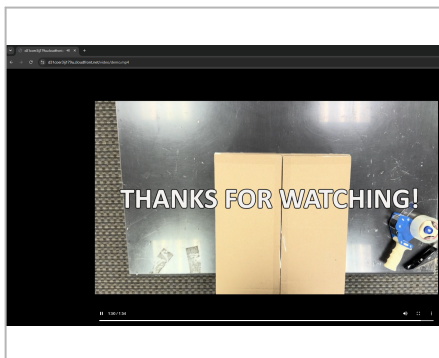
- **Proven end-to-end:** Verified live. Authorized users reach the video automatically, and unauthorized or nonexistent users are cleanly denied.
- **Zero-trust by default:** Even someone with the direct file link cannot access content without valid, signed authorization.
- **Cost:** Built and fully tested within AWS free tier limits, under \$2 total.
- **No vendor lock-in:** Replaces the need for a paid private-video platform with an internally owned, auditable pattern.
- **Reusable pattern:** Directly applicable to internal training videos, restricted SOPs, or any need-to-know media distribution.
- **Real troubleshooting depth:** Diagnosed and resolved production-style issues, including an RSA key-format mismatch and a cookie-scoping bug affecting signature validation.

Stack: Amazon S3 · CloudFront (OAC, Trusted Key Groups) · Cognito · Lambda (Python) · API Gateway · Secrets Manager · IAM · AWS CLI

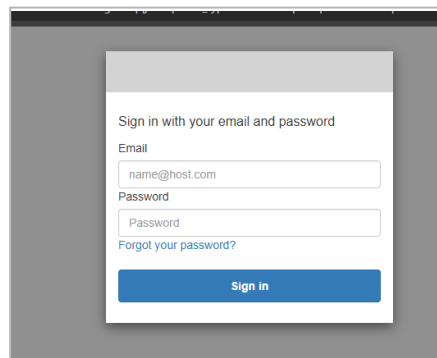
Working Demo: Screenshots



One click entry point to the video



Authorized playback, video streaming live



Secure Cognito sign in screen

Additional projects, including this one's full source and build documentation, available on GitHub: github.com/Taylor-Drummer